



Integrative tensile prediction and parametric analysis of unidirectional carbon/basalt hybrid fiber reinforced polymer composites by bundle-based modeling

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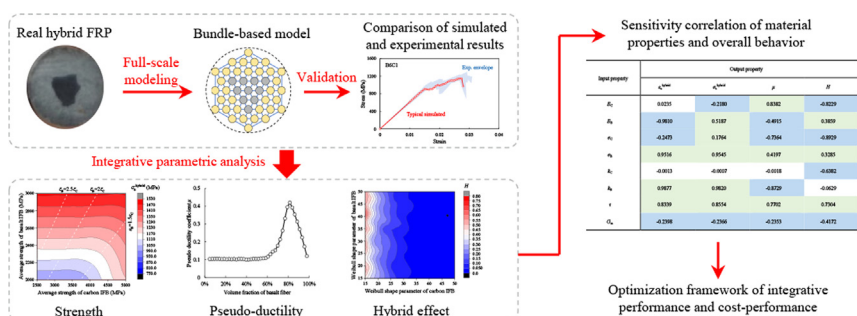
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HIGHLIGHTS

- The tensile behavior of unidirectional hybrid fiber reinforced polymer composites was predicted in a full-scale manner.
- The correlations of model parameters and critical overall properties were revealed by a comprehensive parametric study.
- An optimization framework reflecting the cost-performance efficacy of hybrids was developed based on sensitivities of parameters.

GRAPHICAL ABSTRACT



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ABSTRACT

In this study, the previous bundle-based concept, in which the impregnated fiber bundle is considered as a basic element, was adopted to model unidirectional hybrid fiber-reinforced polymer (FRP) composites in a full-scale manner. Hybrid composites with carbon and basalt fibers were modeled and validated using previous experimental data. A comprehensive parametric study of the constituent properties was performed to determine their sensitivities to overall behaviors. On this basis, an optimization framework was developed to reflect the integrative performance and cost-performance of hybrids. The simulation obtained a saturation point of the matrix shear strength of approximately 70 MPa, beyond which the overall behavior rarely changed. A dispersed arrangement of the fiber bundles could significantly delay the damage propagation and enhance the overall behavior only at low contents of high-elongation fibers. In the present case, the mean ultimate strength can be improved from 1063 MPa to 1315 MPa. The sensitivity analysis indicated that for the elastic moduli of fibers, carbon fiber strength, and Weibull shape parameter of basalt fibers, trade-offs must be considered to balance the overall ultimate strain, strength, ductility, and hybrid effect. The bundle-based modeling concept provides a fast and convenient approach to evaluate the global behavior of hybrids.

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1. Introduction

High strength, light weight, and excellent durability promote the use of fiber-reinforced polymers (FRPs) as high-performance

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materials in civil, aerospace, energy, and automotive engineering [1–4]. However, the significant brittleness of single-type FRP materials causes catastrophic failure at the ultimate tensile state and decreases its design strength. The hybridization of low-elongation (LE) and high-elongation (HE) fibers is a feasible approach to solving this problem, in addition to the direct use of ductile fibers or matrices [5]. The financial costs of pure LE fibers, which in most cases is carbon fiber, can also be reduced through hybridization with HE fibers, such as glass and basalt fibers [6].

In unidirectional hybrid FRP composites, the overall elastic modulus always follows a linear rule of mixture because it is only influenced by the linear ascending stage in the stress–strain curve. Under tensile loads, the LE sub-composite fractures in advance; hence, the overall strength of the hybrid FRP follows a bilinear rule-of-mixture [7]. With the addition of HE fibers, the enhancement of the primary failure strain of LE fibers is known as the hybrid effect, and this phenomenon has attracted significant attention. This phenomenon is attributed to three factors: the residual thermal stress after curing, dynamic stress concentration around the broken fibers, and changes in the failure process [8]. Early experiments performed by Manders and Bader [9] and recent simulations by Guerrero et al. [10] suggested limited effects of the residual stress and dynamic stress concentration. Changes in the failure process play a dominant role in the hybrid effect and other properties [8]. For example, Swolfs et al. [11] conducted a finite element study and found that the stress redistribution was altered when carbon FRP was hybridized with glass fiber.

Efforts have been made to simulate hybrid FRP composites over the past several decades to reasonably feature the aforementioned mechanisms and predict composite behavior. Shear lag (SL) and global load-sharing (GLS) theories are crucial, as they model the hybrids in terms of local and global aspects, respectively. SL models assume that the axial tensile load is resisted by only the fibers, and the matrix only bears the shear load. Zweben first predicted the behavior of hybrid FRP composites using an SL model [12]. Improvements have been made by other researchers to obtain an accurate stress concentration factor around a broken fiber and reliably simulate the failure process [13–15]. For SL models, computational difficulty emerges with an increasing number of fibers, mainly when the statistical strength of fibers is considered. When a representative element is used, determining its dimensions is challenging [16].

The GLS models proposed by Curtin assume that the coplanar fibers transmit an equal axial load after stress redistribution [17,18]. Thus, the GLS models ignore the local effect in the cross-section and minimally reflect the entire damage propagation. Swolfs et al. [8] used this concept to predict the hybrid effect of the failure strain in carbon–glass hybrid FRPs. Rajan and Curtin [6] adopted a similar technique and developed a design method for hybrids.

Recent progress in computer applications facilitates the precise modeling of microscale and mesoscale structures containing numerous fibers, reflecting complicated effects. Mishnaevsky and Dai [19] simulated the damage evolution of hybrids using a three-dimensional finite element method. The effects of the fiber arrangement and constituent properties were investigated. Swolfs et al. [20] observed that fiber dispersion significantly influenced the hybrid effect. The random dispersion of fibers was not the optimal dispersion for maximizing the hybrid effect in this study. A recent study by Guerrero et al. [21] demonstrates that increasing the length or cross-section decreases ductility but increases the hybrid effect, as the number of fibers with random strength also increases.

The bundle-based SL modeling technique, in which an impregnated fiber bundle (IFB), instead of a single fiber, is used as a basic element, was developed by the authors. The entire bundle (yarn) is

impregnated with polymer matrix, and each individual fiber is to be impregnated. Each IFB element can be regarded as a micro-composite. This method originated from SL models [22,23] and was validated by the authors using different FRP composites [24,25]. In comparison with single fibers, it is simpler to evaluate IFB specimens and obtain reliable statistical strength data. This model can be solved numerically with high computational efficiency because of the limited number of IFB elements in a full-scale specimen.

From a practical perspective, the fiber-by-fiber hybridization is still challenging, particularly for mass production. The bundle-by-bundle (tow-by-tow) or layer-by-layer hybrid technique is more technically feasible and financially cost-effective than the fiber-by-fiber method. In this context, the bundle-based modeling concept was exploited in modeling of hybrid carbon/basalt FRP composites in this study. The sensitivities of different model parameters were determined through parametric analysis. Subsequently, a design-oriented optimization framework was developed, which reflected the integrative performance and cost-performance ratio of the hybrid composite.

2. Methodology

2.1. Basic formulations

With bundle-based modeling concept, the hybrid FRP composite is separated into longitudinal IFB elements and transverse matrix elements, shown in Fig. 1. The number of fiber bundle is M and each fiber bundle is segmented into N IFB elements. Therefore, the total number of nodes $N_n = M \times (N + 1)$. For the case in Fig. 1, $M = 7$ and $N = 2$. The length of IFB element is L , and the center-to-center distance of adjacent IFB elements is d , which equals to the length of matrix element.

Considering three-dimensional SL method and assuming a hexagonal packing of IFBs, micromechanical equilibrium equation for a given IFB node (m, n, i) is:

$$A_{ifb} \frac{d\sigma_{m,n,i}(x)}{dx} = -\frac{\pi r_{ifb}}{3} \sum_{l=1}^p \tau_{(m,n),(m',n'),i}^l(x), 1 \leq i \leq N+1 \quad (1)$$

where A_{ifb} and r_{ifb} denote the cross-sectional area and the radius of IFB element, respectively. For hybrid FRP, A_{ifb} and r_{ifb} are not constant in all IFB elements, but depend on the geometrical configuration of IFBs. p is the total number of nodes surrounding the node (m, n, i) . $\sigma_{m,n,i}(x)$ is the axial stress of IFB element (m, n, i) . $\tau_{(m,n),(m',n'),i}(x)$ is the shear stress for the neighboring node (m, n, i) and (m', n', i) . Then, through finite difference method, Eq. (1) can be written as:

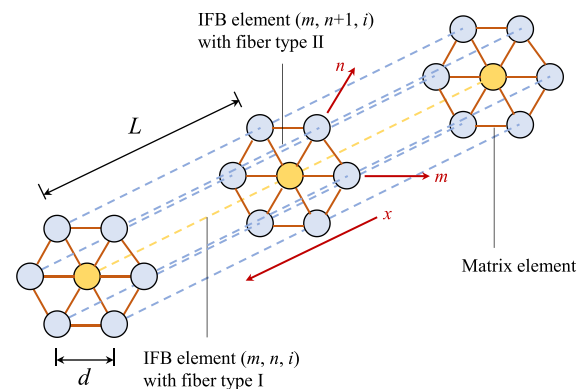


Fig. 1. Analytical model of hybrid FRP composite (m, n = planar axis, x = longitudinal axis).

$$\begin{aligned}
& \frac{E_{ifb} A_{ifb}}{L^2} \left\{ \xi_{m,n,i+1} [u_{m,n,i+1}(x) - u_{m,n,i}(x)] - \xi_{m,n,i} [u_{m,n,i}(x) - u_{m,n,i-1}(x)] \right\} \\
& + \frac{\pi r_{ifb}}{3} \left\{ \psi_{(m,n),(m',n'),i} \frac{G_m [u_{m',n',i}(x) - u_{m,n,i}(x)]}{\delta} + [1 - \psi_{(m,n),(m',n'),i}] \tau_{res} \right\} \\
& = 0, 2 \leq i \leq N
\end{aligned} \quad (2)$$

in which ξ and ψ are damage mark for corresponding IFB and matrix element, respectively. The initial value for ξ and ψ is 1. When the corresponding element is failed, the ξ or ψ is set to 0. E_{ifb} and G_m represent tensile modulus of IFB element and shear modulus of matrix element, respectively. τ_{res} is the residual interfacial shear stress after matrix fails. The boundary conditions are:

$$u_{m,n,i}(x=0) = 0, u_{m,n,N+1}(x=NL) = \varepsilon_{app} NL \quad (3)$$

where ε_{app} is the applied global strain. Then, by combining the finite difference equations Eq. (2) for all nodes, the problem can be expressed as:

$$[K]_{N_n \times N_n} [U]_{N_n \times 1} = [B]_{N_n \times 1} \quad (4)$$

in which $[K]$ is the coefficient matrix of the nodal displacement array $[U]$, and $[B]$ is the array of residual constant terms. By iteration, the nodal displacements can be solved and the stresses can be obtained. The entire calculation terminates at $\det(K) = 0$, which indicates complete failure of IFB elements in a cross-section. The full-range flowchart is shown in Fig. 2, and the readers can find the detailed micromechanical equations and solution process in references [24] and [25].

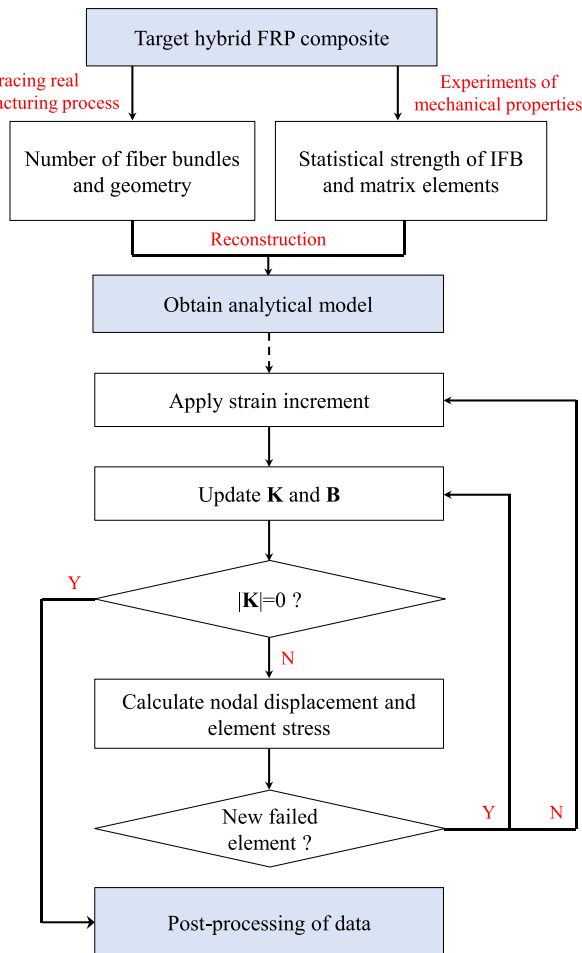


Fig. 2. Full-range flowchart of bundle-based modeling strategy.

2.2. Geometry and parameters

In the reference [26] the authors tested various hybrid FRP tendons with different resin types, fiber fractions, and fiber arrangements. In this study, a part of these tendons were reconstructed using the bundle-based concept. The 6 mm diameter hybrid FRP tendon comprised HE basalt fibers, LE carbon fibers, and a vinyl ester resin matrix, denoted as a B/CFRP tendon. Based on the regular manufacturing process, the fineness of basalt IFB was 1200 tex, and that of carbon IFB was 445 tex [27]. If the fineness of the IFB and the density of the material are known, the equivalent radius (r_{ifb}) and cross-sectional area of the IFB (A_{ifb}) can be obtained. All IFBs shared the same center-to-center distance, i.e., $d = 0.8$ mm. The IFB amount and geometry depend on the actual conditions. Table 1 lists the main material parameters of the B/CFRP hybrid tendons. The statistical strength of the IFB was considered, and could be fitted from existing experimental data. Due to the saturation of imperfections, the lower bound strengths (σ_{LB}) of the basalt and carbon IFBs were set to be 2000 and 2800 MPa, respectively. These lower bound strengths correspond to survival probability of over 99% [24]. The shear strength of the vinyl ester resin (τ) was regarded as a deterministic variable, and a residual stress of $\tau/2$ was selected to determine the effect of interfacial debonding beyond the shear strength of the matrix [28].

Table 2 presents the cross-sectional details of the reconstructed hybrid FRP tendons. In all three types of tendons, the carbon fibers were positioned in the central area. Based on the hexagonal and axisymmetric packing rule, the assignment of IFBs was based on the hybrid volume ratio of basalt and carbon fibers (V_B/V_C). Next, the total fiber volume fraction V_f was re-examined to ensure the validity of the IFB geometry. After trials and verification, the hybrid volume ratios and fiber volume fractions of the reconstructed tendons had relative errors lower than 7%. The most reliable solution is to trace the manufacturing process when several possible geometries are encountered.

The overall length of the tendon was 400 mm, and the entire tendon was longitudinally partitioned into 20 segments. The segment length equaled to the gauge length of element strength in Table 1. Thus, the total number of IFB elements was 800 or 860, depending on the tendon type. For each step, the stresses of the IFB and matrix were calculated using the finite difference method. The failed elements were then identified and marked. When a segment containing IFB elements failed, the calculation was terminated. The strain step for loading was 0.005, and the calculation was performed using the Matlab R2019b platform.

3. Results and discussions

3.1. Validation of results

Fig. 3 shows a comparison between the simulated and experimental tensile stress-strain curves [26]. The typical results obtained using the bundle-based model were in good agreement with the results from tensile tests. As the basalt-to-carbon hybrid volume ratio increased, the overall elastic modulus decreased slightly, and the ultimate tensile strain increased significantly, exhibiting pseudo-ductility behavior. With low volume fractions of LE fibers (Groups B6C1 and B8C1), the remaining basalt IFBs could still sustain the external tensile load, although the curves slightly declined when the carbon IFB started to fracture. Therefore, a secondary stiffness was observed for these groups, and the overall stress still increased with fluctuation, increasing the strength of the hybrid FRP tendons.

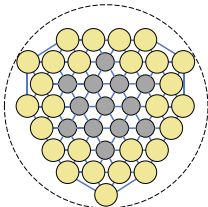
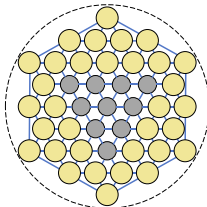
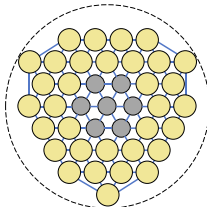
Based on the existing literatures, some critical variables of hybrid FRP are defined in the Appendix. Since the stress-strain

Table 1
Main parameters of B/CFRP hybrid tendons.

Material	Properties					
Carbon IFB	E_{ifb} (GPa)	210	A_{ifb} (mm ²)	0.25	r_{ifb} (mm)	0.38
	Scale (MPa)	3400	Shape	30.0	σ_{LB} (MPa)	2800
Basalt IFB	E_{ifb} (GPa)	75	A_{ifb} (mm ²)	0.45	r_{ifb} (mm)	0.28
	Scale (MPa)	2600	Shape	20.0	σ_{LB} (MPa)	2000
Vinyl ester resin	G_m (GPa)	5.0	τ (MPa)	50.0	τ_{res} (MPa)	$\tau/2$

Note: Scale and shape = Weibull parameters of tensile strength, σ_{LB} = lower bound of tensile strength, G_m = shear modulus, τ = shear strength, and τ_{res} = residual interfacial stress. The IFB elements were assumed to be linearly elastic and the gauge length is 20 mm.

Table 2
Details of reconstructed hybrid FRP tendons.

Property	Group ID		
	B4C1	B6C1	B8C1
IFB geometry			
	$N_B:N_C = 27/13$	$N_B:N_C = 33/10$	$N_B:N_C = 7/33$
Nominal V_B/V_C	4.0	6.0	8.0
Tested V_B/V_C	3.6	6.5	8.2
Reconstructed V_B/V_C	3.8	6.1	8.6
(Relative error)	(+5.6%)	(-6.2%)	(+4.9%)
Tested V_f	58.0%	59.0%	57.5%
Reconstructed V_f	54.5%	61.5%	58.9%
(Relative error)	(-6.0%)	(+4.2%)	(+2.4%)

Note: N_B , N_C = amounts of basalt IFBs and carbon IFBs, respectively.

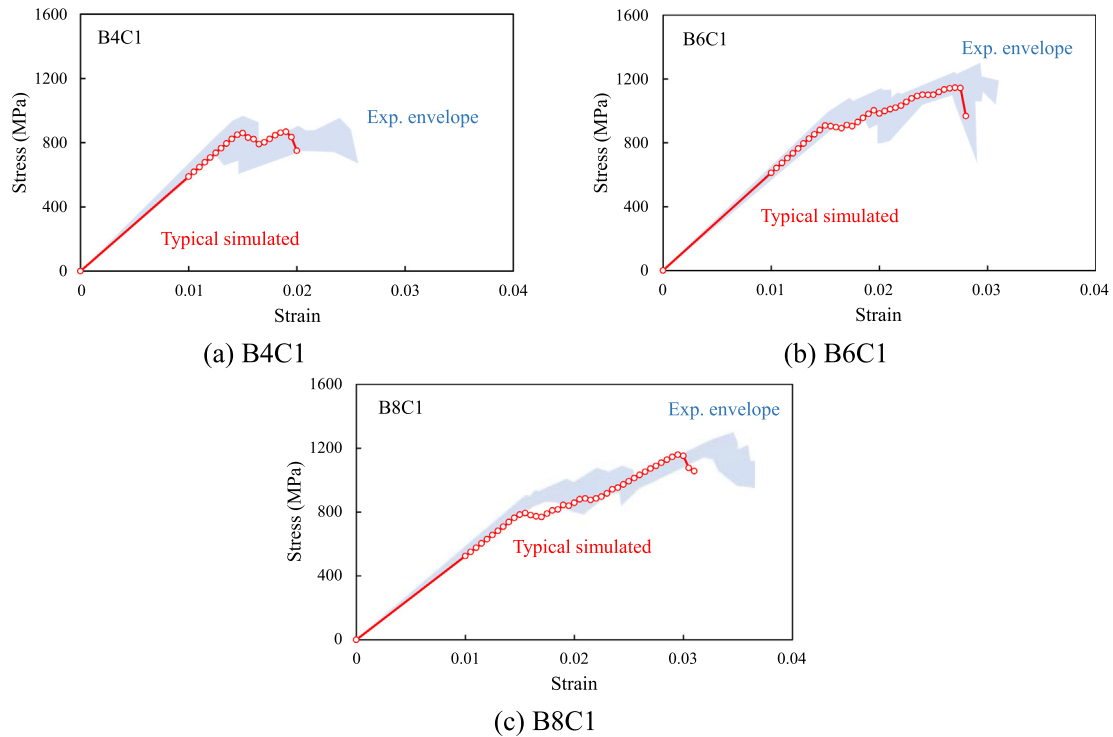


Fig. 3. Simulated and experimental stress-strain curves (experimental data obtained from reference [26]).

curve can be obtained for each simulation, those important variables can be captured from the simulated results. Table 3 lists the average values of the 1000 simulations. For all items, the sim-

ulated results had relative errors of less than $\pm 10\%$. The biases were mainly caused by the inaccuracy of the input statistical IFB strength, interfacial behavior, and other material properties.

Table 3

Results of 1000 simulations and experiments.

Group		σ_u^{hybrid} (MPa)	E^{hybrid} (GPa)	$\epsilon_u^{\text{hybrid}}$
B4C1	Experimental	884	64	0.024
	Simulated	859	59	0.022
	(Relative error)	(−2.8%)	(−7.8%)	(−8.3%)
B6C1	Experimental	1240	60	0.030
	Simulated	1119	58	0.029
	(Relative error)	(−9.7%)	(−3.3%)	(−3.3%)
B8C1	Experimental	1240	55	0.036
	Simulated	1123	53	0.033
	(Relative error)	(−9.4%)	(−3.6%)	(−8.3%)

Note: E^{hybrid} = elastic modulus of hybrid FRP tendon. The experimental data were obtained from reference [26].

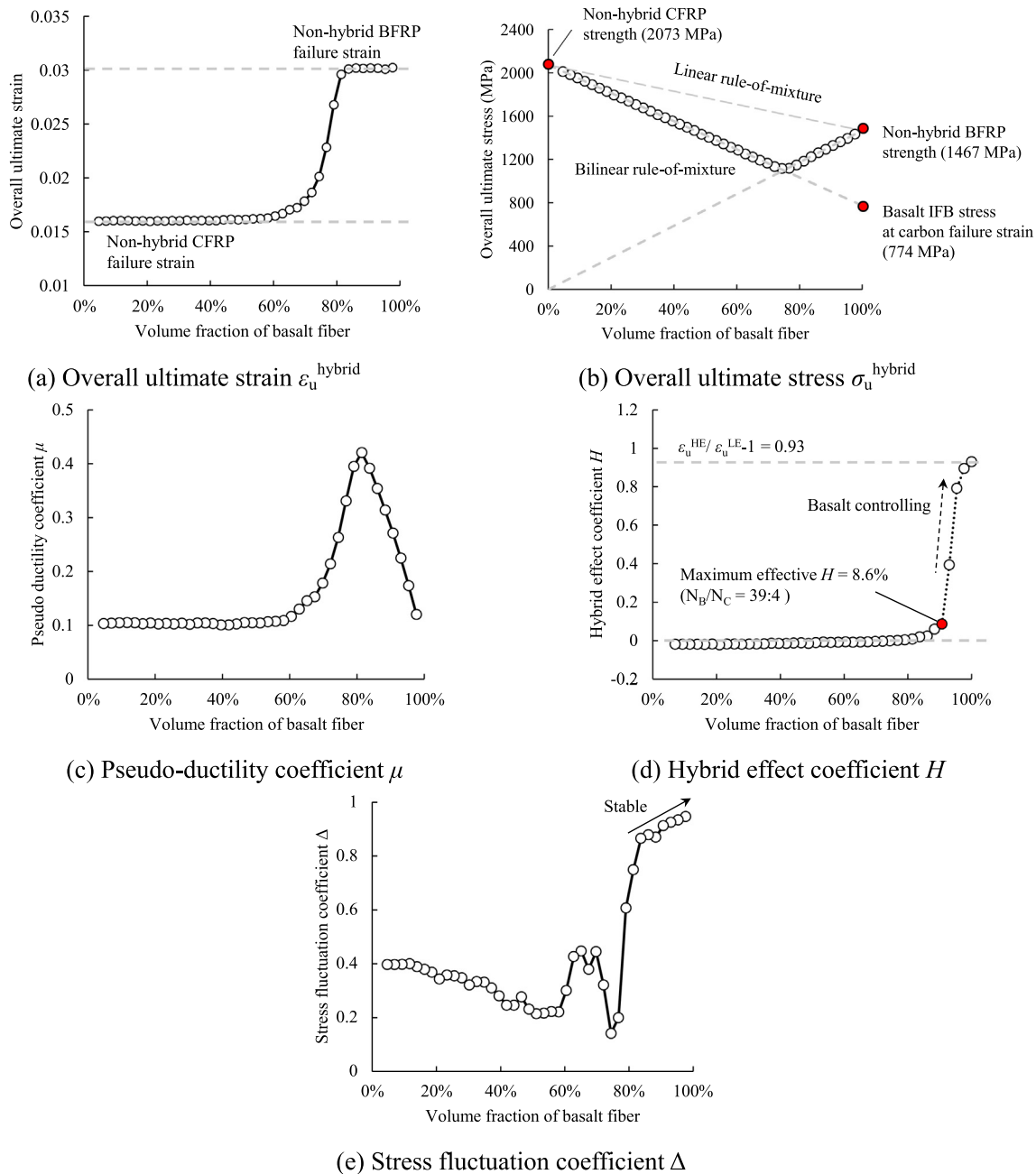


Fig. 4. Average results of 1000 simulations for varying amounts of basalt IFB (see the relevant definitions in Appendix.).

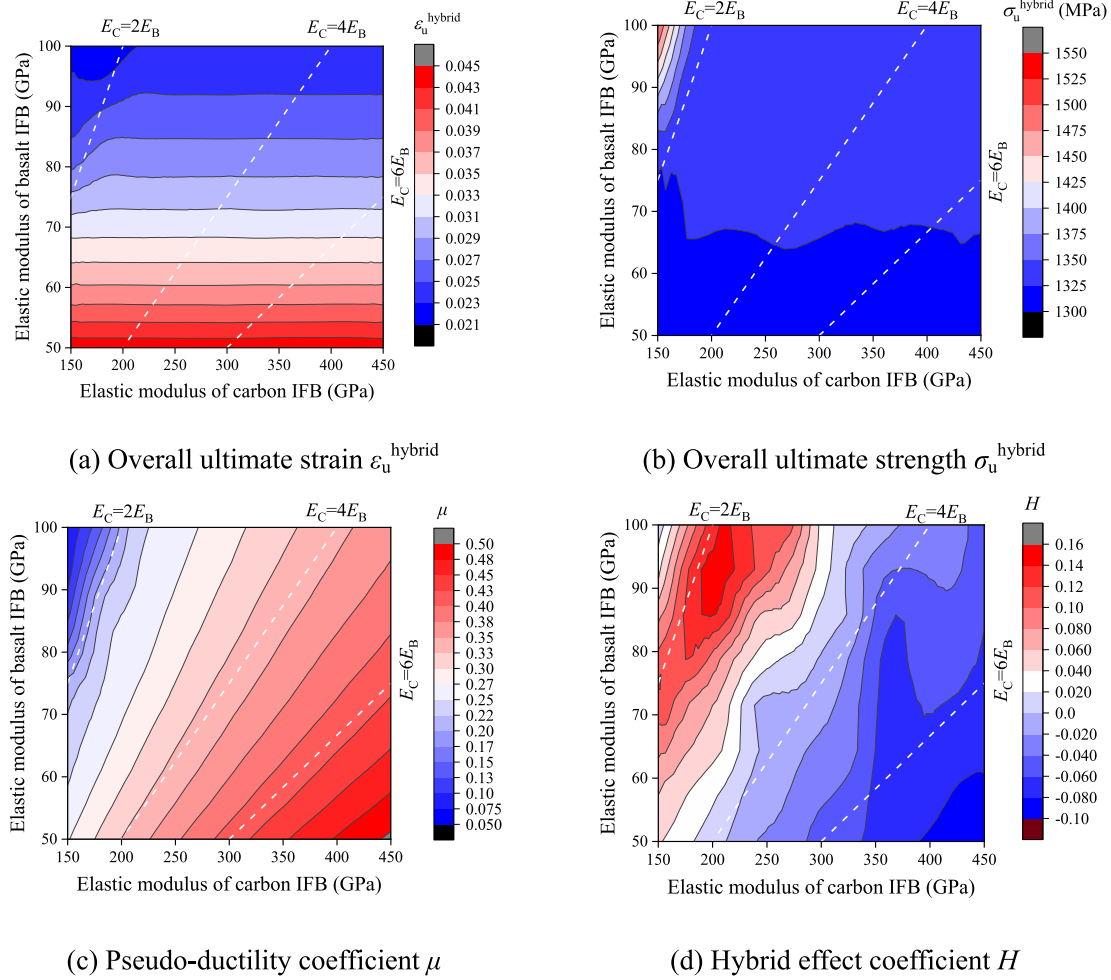


Fig. 5. Effects of IFB elastic modulus (fixed IFB strength).

3.2. Parametric study

After verifying the effectiveness of the proposed bundle-based model, some parameters were investigated, adopting Group B6C1 as the baseline. For simplicity and comparability, the A_{ifb} values of the basalt and carbon fibers were set to be equal (0.45 mm^2) to make the overall fiber content constant ($V_f = 0.69$). With 43 IFBs in total, the average overall ultimate strains of the nonhybrid LE composite ($\varepsilon_u^{\text{LE}}$) and HE composite ($\varepsilon_u^{\text{HE}}$) were simulated to be 0.016 and 0.030, respectively.

3.2.1. Effect of hybrid volume ratio

The relative amounts of LE and HE fibers are significant in hybridization design [5]. In this study, the proportion of the basalt IFB volume to the total fiber volume varied from 0% to 100%, and the total fiber content was fixed. It was observed that the $\varepsilon_u^{\text{hybrid}}$ experienced a gradual variation as the HE fiber volume percentage increased to approximately 60% (Fig. 4(a)), indicating that the basalt IFBs had a dominant influence on the overall failure. The development of σ_u^{hybrid} followed the classical bilinear rule-of-mixture (Fig. 4(b)) [7], and the minimum strength was 1117 MPa at an HE fiber volume fraction of 74% ($V_B/V_C = 32:11$). This point was also the force equilibrium point between the LE and HE sub-composites, obtained as $E_C/(E_C + E_B) = 210/(210 + 75) = 74\%$. In terms of pseudo-ductility (Fig. 4(c)), an optimum HE fiber volume

fraction of 81% ($V_B/V_C = 35:8$) yielded the maximum μ value of 0.42.

Beyond the critical volume fraction of 91% ($V_B/V_C = 39:4$; Fig. 4(d)), the hybrid effect coefficient H sharply increased because the overall failure tended to be entirely controlled by the basalt IFBs. The upper bound of H (the gray dashed line) represents the relative enhancement from the carbon failure strain ($\varepsilon_u^{\text{LE}}$) to the basalt failure strain ($\varepsilon_u^{\text{HE}}$). Typically, the hybrid effect can be regarded as insignificant at this stage (the dashed line); hence, the maximum effective H is 8.6% at the critical volume fraction. This phenomenon is consistent with the findings reported in reference [8]. Although the Weibull shape parameter of carbon IFB strength is large, each IFB contains approximately 12,000 filaments for T800 carbon fiber, and the corresponding Weibull shape parameter of carbon filament strength is much smaller. Thus, the hybrid effect can still be observed. For the stress fluctuation, Δ was positively related to σ_u^{hybrid} (Fig. 4(d)), as Spearman's correlation coefficient between the data points in Fig. 4(b) and Fig. 4(d) reached 0.9380. Beyond an HE fiber volume fraction of 74%, the stability of the fluctuating stage significantly increased with increasing σ_u^{hybrid} .

3.2.2. Effect of IFB properties

In this section, the basalt volume fraction was fixed to 91% ($V_B/V_C = 39:4$), which exhibited the maximum effective H , as shown in Fig. 4(d). The material properties of interest were then varied with the other fixed parameters. Before performing the simulations, the

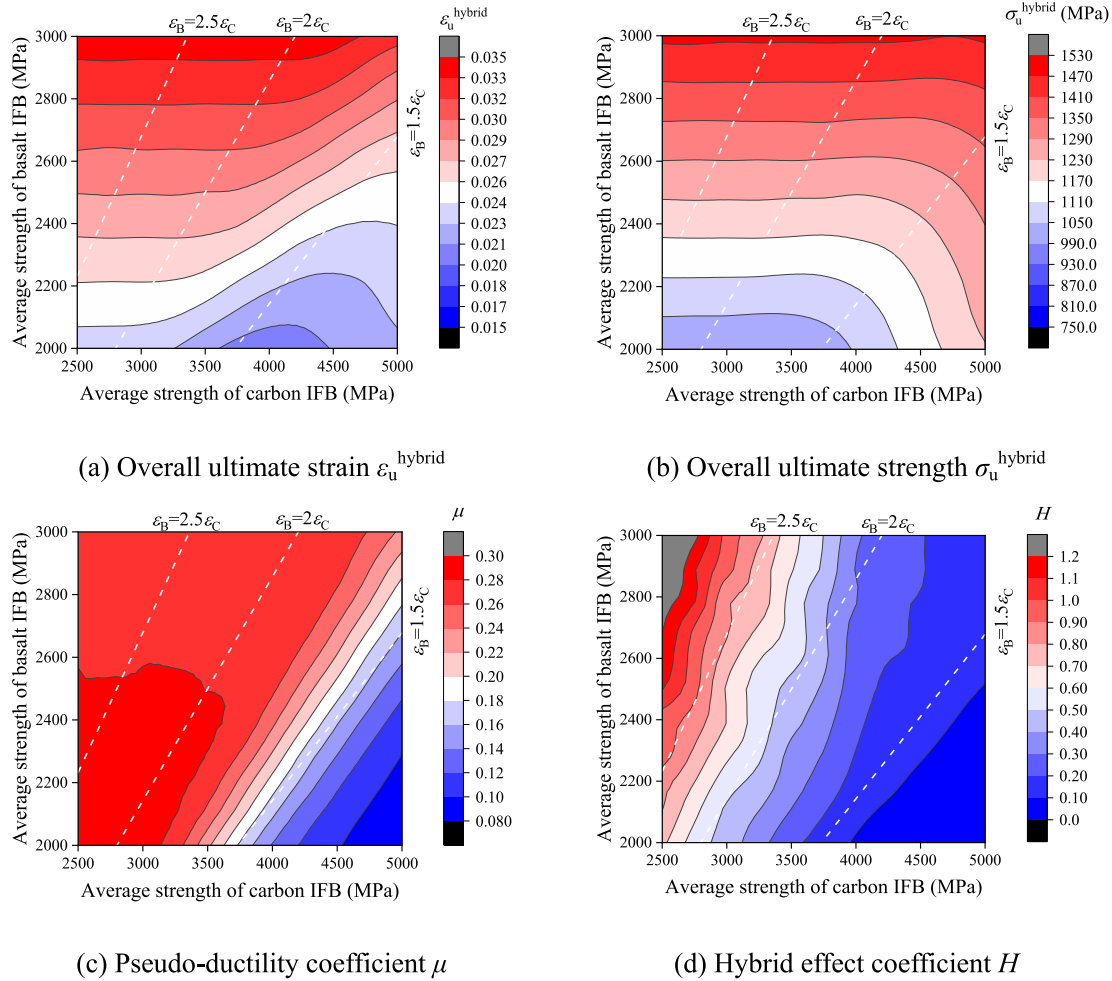


Fig. 6. Effects of IFB strength (fixed IFB elastic modulus; ε_B and ε_C are the average failure strains of basalt and carbon IFBs, respectively).

average overall ultimate strain of the nonhybrid LE composite ($\varepsilon_u^{\text{LE}}$) was calculated to determine the hybrid effect coefficient H because $\varepsilon_u^{\text{LE}}$ varied with the input parameters.

Figs. 5–7 depict the correlation between the IFB and overall properties. At fixed strengths, $\varepsilon_u^{\text{hybrid}}$ was mainly influenced by the basalt IFB modulus (Fig. 5(a)). σ_u^{hybrid} was somewhat insensitive to the IFB moduli, except when the basalt IFB modulus was high (Fig. 5(b)) because the carbon and basalt elements fractured simultaneously in these circumstances. Fig. 5(c) shows that brittle carbon IFBs and ductile basalt IFBs can reach the optimum pseudo-ductility of the hybrid FRP composite. μ hardly varied with a constant modulus ratio E_C/E_B (or failure strain ratio). The hybrid effect coefficient H (Fig. 5(d)) exhibited an opposite trend compared to that of the ductility. The maximum H value was 0.16 at $E_C = 210$ GPa and $E_B = 95$ GPa. In addition, a higher modulus ratio E_C/E_B (white dashed lines) resulted in better ductility but a less significant hybrid effect.

Fig. 6 sustained the normal strength distributions for both IFB types and selected a coefficient of variation of 10%. At fixed IFB elastic moduli, $\varepsilon_u^{\text{hybrid}}$ and σ_u^{hybrid} were mainly influenced by the basalt strength. The carbon strength influenced the results only when it reached a threshold, at which the failure strains of the carbon and basalt IFBs overlapped, for instance, at $\varepsilon_B = 1.5\varepsilon_C$. It can be inferred from Fig. 6(c) and (d) that a higher failure strain ratio $\varepsilon_B/\varepsilon_C$ (or strength ratio) results in higher ductility and induces a more significant hybrid effect. Similar findings were reported in [29]. Although the pseudo-ductile strain ε_{pd}

increased with increasing $\varepsilon_B/\varepsilon_C$, the ratio of ε_{pd} to $\varepsilon_u^{\text{hybrid}}$ (i.e., μ) almost plateaued at a critical $\varepsilon_B/\varepsilon_C$ (for example, $\varepsilon_B = 2.5\varepsilon_C$ in Fig. 6(c)).

The scatter of basalt strength influenced $\varepsilon_u^{\text{hybrid}}$, σ_u^{hybrid} , and μ , whereas that of carbon strength significantly influenced the hybrid effect (Fig. 7). It was anticipated that at a sufficiently high Weibull shape parameter of the carbon IFB, the hybrid effect would disappear. Previous studies also showed that a high strength scatter of the LE fibers improved the hybrid effect [8,30].

3.2.3. Effect of matrix properties

In Fig. 8, the effect of the matrix (interfacial) properties was evaluated. Beyond the critical shear strength of 70 MPa, $\varepsilon_u^{\text{hybrid}}$, σ_u^{hybrid} , and μ remained almost unchanged because the interface was sufficiently strong to transfer the axial load from the broken elements, and no matrix element failed. The relationship between the critical strength value and other parameters should be determined in the future to enhance the compatibility of the matrix. In addition, a tough matrix (low shear modulus) was beneficial to the overall properties. This result is rather consistent with the findings of Na et al. In their work, an optimal interfacial shear strength of 70 MPa was found in carbon fiber-epoxy composites to reach the highest tensile strength [31].

3.2.4. Effect of hybrid IFB arrangements

The fiber arrangement in hybrid composites has attracted significant attention. When $V_B/V_C = 30/13$, the dispersed packing of

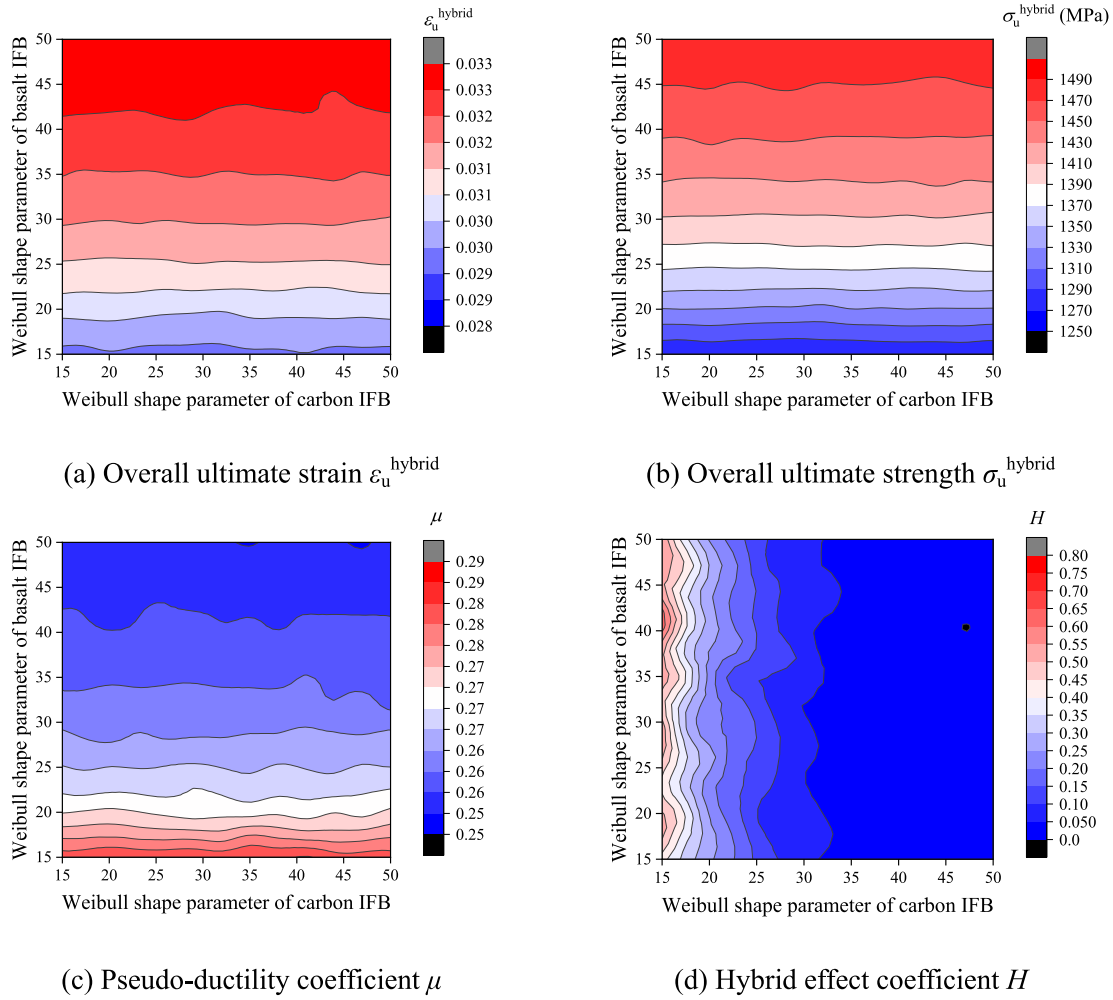


Fig. 7. Effects of Weibull modulus of IFB strength (fixed Weibull scale parameter).

the carbon IFBs did not influence the overall stress–strain curves, compared to centralized packing (Fig. 9(a) and (b)). However, the average $\varepsilon_u^{\text{hybrid}}$ significantly increased by 31% in the dispersed model at $V_B/V_C = 39/4$ (In Fig. 9(c) and (d)). Furthermore, the cross-sectional stress profiles at the initial damage strain was shown in Fig. 10. The centralized packing increased the stress concentration of the basalt IFBs near the broken carbon IFBs, which resulted from the local load-sharing rule of the model. However, because of the limited number of IFB elements in the cross-section, the local stress concentration was accompanied by global stress enhancement. Thus, combined “local” and “global” effects were observed in the bundle-based model. As mentioned earlier, a force equilibrium point existed between the LE and HE subcomposites ($V_B/V_C = 32/11 = 2.9$ in this case). When $V_B/V_C < 32/11$ (for instance, $V_B/V_C = 30/13$), the positive influence of dispersion disappeared because the global stress enhancement attributed to the broken carbon IFBs was significant, and the remaining basalt IFBs fractured abruptly. In contrast, when $V_B/V_C > 32/11$ (for instance, $V_B/V_C = 39/4$), the basalt IFBs had the potential to attain their failure strains, indicating the influence of the dispersion. This mechanism was quantitatively demonstrated in Fig. 11. The centralized model underwent a rapid damage process and failed prematurely. However, the dispersed model fully utilized the carbon and basalt IFBs. Nageh et al. [26] investigated B/CFRP hybrid tendons with V_B/V_C values of 4 and 6, which exceeded the threshold $V_B/V_C = 2.9$. Thus, the experimental dis-

persed tendons exhibited better overall behavior than that of the centralized tendons.

Table 4 further summarized the simulated overall properties of the groups involved. With $V_B/V_C = 30/13$, a slight negative hybrid effect coefficient H can be observed both for centralized and dispersed packings. With a lower carbon fiber content, the H increased from 5.30% to 7.91% when using dispersed packing instead of centralized one. The pseudo-ductility was also significantly enhanced with dispersed packing.

3.3. Limitations of present study

Despite of advantages of the bundle-based modeling concept and findings in this study, several limitations should be remarked:

- (1) The bundle-based modeling method starts from IFB elements, which integrates number of filaments inside the IFB. Therefore, the very local behavior at filament level is overlooked, and this study mainly focus on prediction of overall tensile behaviors at full-scale level. Essentially, the bundle-based modeling concept is an equivalent modeling strategy to pursue a higher computation efficiency.
- (2) Some model parameters require further careful determinations. For instance, the center-to-center distance of IFBs and residual ratio of interfacial shear stress, can be obtained from real experiments.

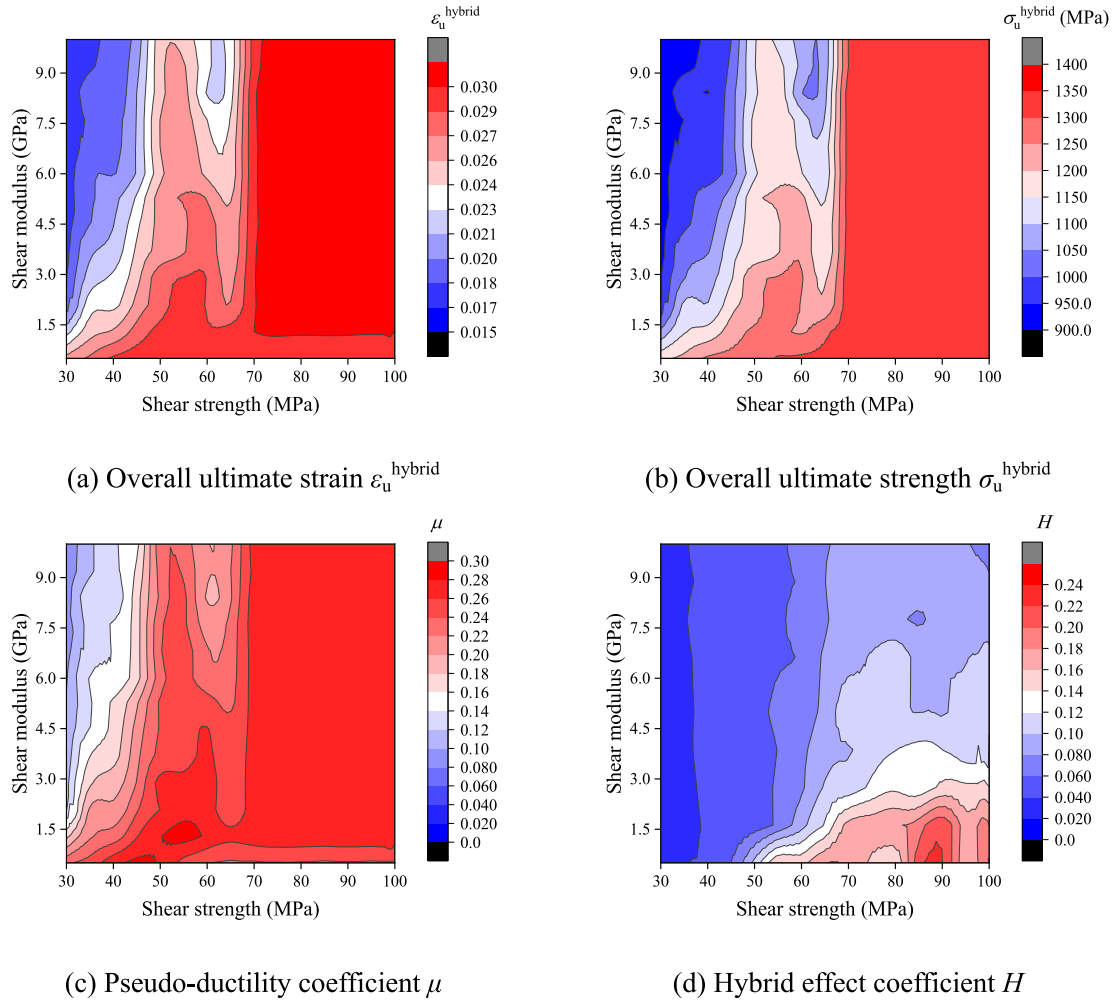


Fig. 8. Effects of matrix (interfacial) shear strength and modulus (residual stress ratio = 0.5).

- (3) With HE fiber volume ratio increasing, the growth of hybrid effect coefficient H can be influenced by the size effect of LE sub-composite, because the absolute amount of LE fibers was reduced, leading to a higher rupture strain of LE sub-composite. The size effect is not considered in this study, and deserves further investigation.

4. Integrative design

4.1. Sensitivity of parameters on overall properties

The sensitivity analysis of the main parameters provided the priority ranking for further optimization. The standardized regression coefficient (SRC) was used to eliminate the gap between scales in different variables [32]. When $-1 < \text{SRC} < 1$, a higher absolute SRC value indicates greater sensitivity. The plus and minus signs of the SRC indicate positive and negative effects, respectively. In this study, the input data in Section 3. Parametric study and regression were performed using SPSS.

As listed in Table 5, the orders of the SRCs were as follows: for $\varepsilon_u^{\text{hybrid}}$, $k_B > \sigma_B > \tau \gg E_C \approx 0$, and $E_B < \sigma_C < G_m \ll k_C \approx 0$; for σ_u^{hybrid} , $k_B > \sigma_B > \tau > E_B > \sigma_C > 0$, and $G_m < E_C < k_C \approx 0$; for μ , $E_C > \tau > \sigma_B > 0$, and $k_B < \sigma_C < E_B < G_m \ll k_C \approx 0$; for H , $\tau > E_B > \sigma_B > 0$, and $\sigma_C < E_C < k_C < G_m < 0$. For all output properties, the absolute strength of the HE fiber (σ_B) and shear strength of the matrix (τ) exhibited positive effects, whereas an increased shear modulus of the matrix (G_m) was disadvantageous. A lower k_C , which indicated a larger scatter

of strength of the LE fiber, proved to be beneficial to the hybrid effect but was negligible for other properties. For other input properties, including E_C , E_B , σ_C , and k_B , trade-offs must be considered to balance the output properties.

4.2. Cost-performance index

The priority of the output properties ($\varepsilon_u^{\text{hybrid}}$, σ_u^{hybrid} , μ , and H) varies in the practical design of structures. Moreover, the cost issue is critical when a large quantity of raw materials is used; this has rarely been considered previously. In general, the price of a fiber bundle (or tow) increases with its strength and modulus and decreases with its strength scatter. Thus, ignoring the contribution of the matrix, an empirical expression of total financial cost of a hybrid FRP composite may express as follows:

$$C^{\text{hybrid}} = N_C \alpha_1 \left(\frac{E_C}{E_C^{\text{ref}}} \right)^{\beta_1} \left(\frac{\sigma_C}{\sigma_C^{\text{ref}}} \right)^{\beta_2} \left(\frac{k_C}{k_C^{\text{ref}}} \right)^{\beta_3} + N_B \alpha_2 \left(\frac{E_B}{E_B^{\text{ref}}} \right)^{\beta_4} \left(\frac{\sigma_B}{\sigma_B^{\text{ref}}} \right)^{\beta_5} \left(\frac{k_B}{k_B^{\text{ref}}} \right)^{\beta_6} \quad (5)$$

where E_C^{ref} , σ_C^{ref} , k_C^{ref} , E_B^{ref} , σ_B^{ref} , and k_B^{ref} are the reference input properties of the corresponding parameters in the above expression. N_C and N_B are the numbers of the carbon and basalt IFBs, respectively, and α_1 , α_2 , β_1 , and β_6 are constants. The integrative performance P^{hybrid} can be expressed as follows:

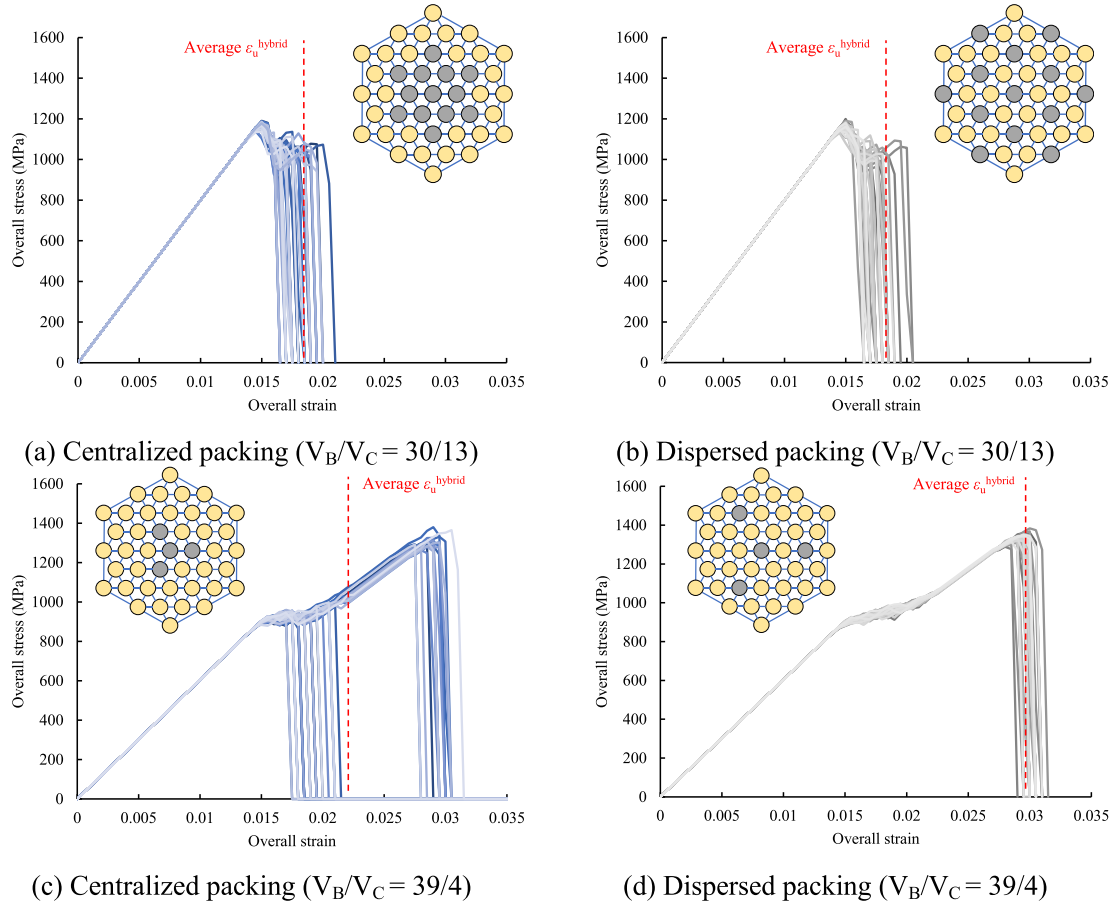


Fig. 9. Overall stress–strain curves with different carbon IFB packings (50 simulations).

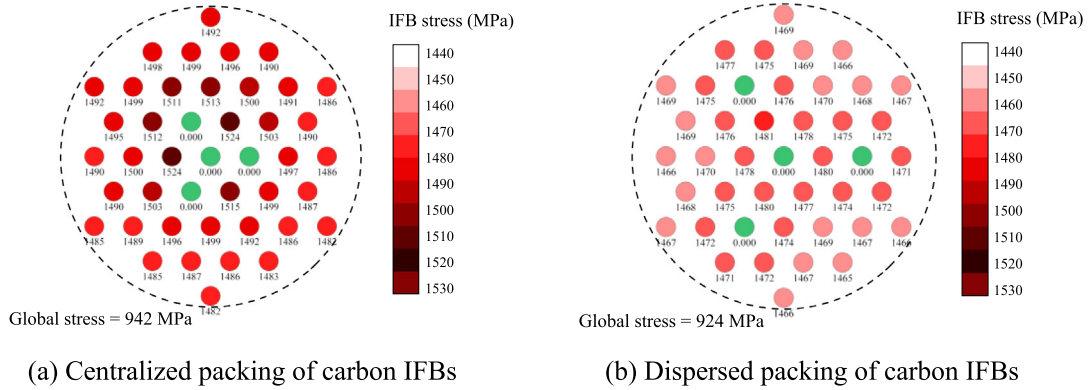


Fig. 10. Cross-sectional stress profiles for initial damage strains of carbon IFBs ($N_B/N_C = 39/4$).

Table 4

Average overall properties of carbon centralized and dispersed cases (1000 simulations).

Group	Output property			
	σ_u^{hybrid}	σ_u^{hybrid}	μ	H
$V_B/V_C = 30/13$, centralized	0.0171	1166	0.1427	−0.0167
$V_B/V_C = 30/13$, dispersed	0.0167	1166	0.1224	−0.0170
$V_B/V_C = 39/4$, centralized	0.0225	1063	0.2071	0.0530
$V_B/V_C = 39/4$, dispersed	0.0297	1315	0.2648	0.0791

$$p^{\text{hybrid}} = \sum_{i=1}^m w(i) \frac{y_k(i)}{y_0(i)} \quad (6)$$

where $y_k(i)$ indicates the i th output variables for the k th iteration, $w(i)$ denotes the corresponding weighting coefficient used to describe the priority, and m denotes the total number of output variables. Y_0

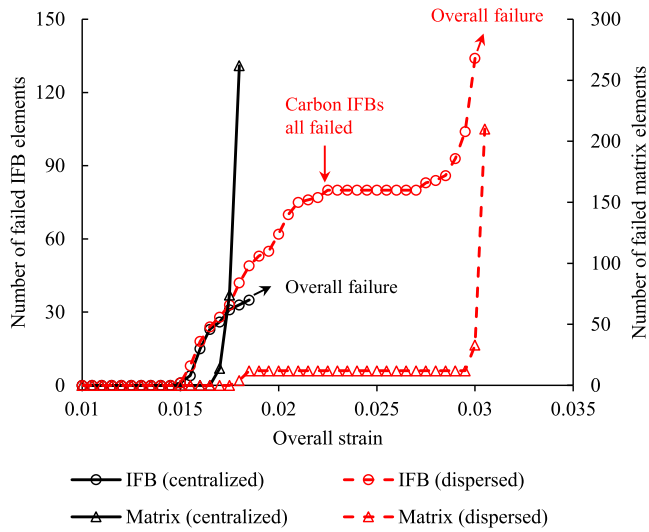


Fig. 11. Variation in overall strain with number of damaged elements ($N_B/N_C = 39/4$).

represents the output array of the initial scenario. Next, the cost-performance index η can be defined as follows:

$$\eta = \frac{P^{\text{hybrid}}}{C^{\text{hybrid}}} \quad (7)$$

4.3. Example

As an example, the model parameters discussed in Section 3. Parametric study was performed, in which $V_B:V_C$ was fixed at 39:4. The overall elastic modulus of the hybrid composite E^{hybrid}

was maintained, and it obeyed the linear rule of mixtures. Thus, Eq. (4) is valid.

$$\left(\frac{4}{43}E_C + \frac{39}{43}E_B\right)V_f = E^{\text{hybrid}} = \text{constant} \quad (8)$$

For simplicity, k_C and k_B are fixed at 30 and 20, respectively, and the total cost equation is expressed in an ideal form, as follows:

$$C^{\text{hybrid}} = 10 \times 4 \left(\frac{E_C}{210}\right)^2 \left(\frac{\sigma_C}{3400}\right) + 39 \left(\frac{E_B}{75}\right)^2 \left(\frac{\sigma_B}{2600}\right) \quad (9)$$

where E_C and E_B are expressed in gigapascals, and σ_C and σ_B are expressed in megapascals. Assume the price of fibers is more sensitive to their moduli than stress. The correct relationship between price and properties requires sufficient marketing research. The weighting array $\mathbf{W}$ was set to {9,9,9,9}. For the initial scenario, $E_C = 210$ GPa, $E_B = 75$ GPa, $\sigma_C = 3400$ MPa, $\sigma_B = 2600$ MPa, and $C^{\text{hybrid}} = 79$. By using the bundle-based model, the following parameters can be obtained: $\mathbf{Y}_0 = [0.0302; 1329 \text{ MPa}; 0.2712, 0.1073]$, initial $P^{\text{hybrid}} = 36$, and $\eta = 18/79 = 0.4556$.

By enumerating the combinations of E_C , σ_C , and σ_B , the optimal combinations for varied values of $\mathbf{W}$ can be obtained (Table 6). For example, at $\mathbf{W} = \{18, 2, 14, 2\}$ ($\epsilon_u^{\text{hybrid}}$ and μ as the design priority), a low σ_C value and a high σ_B value were required to achieve the highest integrative performance P^{hybrid} of 38.2. When considering the optimal cost performance, however, low E_C and σ_B values were suitable; thus, η was increased from 0.4566 to 0.6716, an increase of 47.4%. By varying the weighting array $\mathbf{W}$, namely, the design priority, the optimal combination also changed. The price function in Eq. (9) should be carefully specified, and additional boundaries, such as the minimum overall strain or strength, should be included. When the number of involved parameters increases, a rational optimization algorithm must be adopted to minimize the computational burden.

Table 5
SRCs of input and output properties.

Input property	Output property			
	$\epsilon_u^{\text{hybrid}}$	σ_u^{hybrid}	μ	H
E_C	0.0235	-0.2180	0.8382	-0.8229
E_B	-0.9810	0.5187	-0.4915	0.3859
σ_C	-0.2473	0.1764	-0.7364	-0.8929
σ_B	0.9516	0.9545	0.4197	0.3285
k_C	-0.0013	-0.0007	-0.0018	-0.6382
k_B	0.9877	0.9820	-0.8729	-0.0629
τ	0.8339	0.8554	0.7702	0.7304
G_m	-0.2398	-0.2366	-0.2353	-0.4172

* Note: σ_C , σ_B = average strengths of carbon and basalt IFBs, respectively; k_C , k_B = Weibull shape parameters of carbon and basalt IFB strengths, respectively; τ , G_m = shear strength and shear modulus of the matrix, respectively. The green block indicates a significant negative effect, blue block indicates a significant negative effect, and other blocks indicate a negligible effect.

Table 6
Input and output properties of optimal combinations.

	Weighting array $\mathbf{W}$	Input				Output		
		E_C (GPa)	E_B (GPa)	σ_C (MPa)	σ_B (MPa)	p^{hybrid}	C^{hybrid}	η
Initial	{9,9,9,9}	220	75.0	3400	2600	36	79.0	0.4556
Optimal p^{hybrid}	{18,2,14,2}	220	74.2	2500	2800	38.2	73.4	0.5198
	{12,12,6,6}	150	81.4	2500	3000	42.8	68.0	0.6308
	{6,8,20,2}	220	74.2	3900	3000	39.2	94.4	0.4152
Optimal η	{18,2,14,2}	150	81.4	2500	1500	27.8	41.5	0.6716
	{12,12,6,6}	150	81.4	2600	1500	37.2	42.1	0.8854
	{6,8,20,2}	150	81.4	2500	1500	29.6	41.5	0.7132

5. Conclusions

In this study, an IFB instead of a single fiber was adopted as a basic element to model unidirectional hybrid FRP composites. The effectiveness of the model was first validated using previous experimental data. Next, a systematic parametric study was conducted, and the results were compared with the existing findings. The sensitivity of the input parameters was obtained. Finally, a design-oriented optimization framework that considers the integrative performance and cost-performance index of the hybrids was proposed. The main conclusions of this study are as follows.

- (1) A tough matrix enhanced the overall behavior of the hybrid FRP composite, but a saturation point of the matrix shear strength was observed, beyond which the $\varepsilon_u^{\text{hybrid}}$, σ_u^{hybrid} , and μ rarely changed. In the present study, a critical shear strength of approximately 70 MPa was observed.
- (2) With a low LE fiber content, a dispersed arrangement of the fiber bundles could significantly delay the damage propagation and improve the overall behavior. When $V_B/V_C = 39/4$, a disperse packing of carbon IFBs can improve the mean $\varepsilon_u^{\text{hybrid}}$ from 0.0225 to 0.0297. However, with medium and high contents, the improvement was negligible.
- (3) For input parameters E_C , E_B , σ_C , and k_B , trade-offs must be considered to balance the output properties, such as $\varepsilon_u^{\text{hybrid}}$, σ_u^{hybrid} , μ , and H . The input parameters σ_B and τ always exhibited a positive influence on the output properties.

- (4) Based on the proposed optimization framework, the optimal combinations of input parameters can be obtained to maximize the integrative performance and cost-performance index.

Data availability statement

All data, models, and codes generated or used during this study appear in the paper.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Definitions of critical variables

As shown in Table A.1, the stress-strain curve of the hybrid FRP composite can be divided into three stages. In the early ascending

Table A1
Conceptualized stress-strain curve of hybrid FRP composite and relevant definitions.

Graph

The graph illustrates the stress-strain behavior of a hybrid FRP composite. The curve is divided into three main stages: Ascending, Fluctuating, and Failing. The Ascending stage shows a linear increase in stress with strain. The Fluctuating stage is characterized by multiple peaks and valleys, representing the damage initiation and propagation. The Failing stage shows a sharp drop in stress. Key points on the curve include the origin (ε₀), the first peak (ε₁), the last convex point (ε_Ncv), and the ultimate strain (ε_u^hybrid). A linear fitting line is shown for the fluctuating stage, and a red 'x' marks the damage initiation point (ε_pd). The y-axis also marks σ₀, σ₁, and σ_u^hybrid.

Legend

- Damage initiation point
- Convex point
- Concave point

Notation	Indication	Definition
$\varepsilon_u^{\text{hybrid}}$	Overall ultimate strain	The strain of the last convex point ε_{Ncv}
σ_u^{hybrid}	Overall ultimate strength	The maximum stress of the convex points
μ	Pseudo-ductility coefficient	The ratio of ε_{pd} to $\varepsilon_u^{\text{hybrid}}$
H	Hybrid effect coefficient	The relative enhancement from $\varepsilon_u^{\text{LE}}$ to ε_1
Δ	Stress fluctuation coefficient	The R^2 value for the fitting line in the fluctuating stage

stage, all the elements are intact. After the damage initiation point, the weakest LE IFB element starts to fail, and a progressive fracture of the LE elements may lead to an inelastic stage until the first convex point is attained. Let ε_u^{LE} be the overall ultimate strain of the nonhybrid LE composite. Based on Swolfs et al. [5], the hybrid effect coefficient H is defined here as the enhancement percentage from ε_{LE} to ε_1 . Subsequently, the curve enters a fluctuating stage as the tensile load gradually transfers from the failed LE element to the HE element, and the HE element starts to fail. Multiple convex and concave points can be captured at this stage. The total numbers of the convex and concave points are denoted as N_{cv} and N_{cc} , respectively. In particular, the previous studies rarely quantified the amplitude of stress fluctuation. In this context, define the stress fluctuation coefficient Δ as the R -squared value obtained from the fitting line of the fluctuating stage. A higher Δ value indicates less stress fluctuation, which is beneficial for the practical design of the fluctuating stage. At the last convex point, the overall ultimate strain ε_u^{hybrid} and pseudo-ductility coefficient μ can be determined. It should be noted that the definition of the pseudo-ductile strain ε_{pd} is obtained from the literature [33], μ is the proportion of the pseudo-ductile strain to the ultimate strain. (Table A1).

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